## ■ D

`D[`$f$`, `$x$`]` gives the partial derivative $\frac{\partial}{\partial x} f$.

`D[`$f$`, {`$x$`, `$n$`}]` gives the multiple derivative $\frac{\partial^n}{\partial x^n} f$.

`D[`$f$`, `$x_1$`, `$x_2$`, `…`]` gives $\frac{\partial}{\partial x_1} \frac{\partial}{\partial x_2} \ldots f$.

All quantities that do not explicitly depend on the $x_i$ are taken to have zero partial derivative. ■ `D[`$f$`, `$x_1$`, …, NonConstants -> {`$v_1$`, …}]` specifies that the $v_i$ implicitly depend on the $x_i$, so that they do not have zero partial derivative. ■ The derivatives of built-in mathematical functions are evaluated when possible in terms of other built-in mathematical functions. ■ `D` uses the chain rule to simplify derivatives of unknown functions. ■ See page 412. ■ See also: `Dt`, `Derivative`.

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